ABSTRACT

LASTNAME, FIRSTNAME MIDDLENAME. Title of This Dissertation. (Under the direction of Chair Person).

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These instructions are meant to help you get started gently. You do not need to understand every file in the template before you begin. The main idea is that a thesis or dissertation is too large to keep comfortably in one file, so this project is split into a main file, a front matter file, chapter files, appendix files, a bibliography file, and a few support files.

The main file is `dissertation-main-accessible.tex`. This is the file you compile to create your thesis or dissertation PDF. It controls the document setup, accessibility metadata, packages, front matter, chapters, bibliography, and appendices.

Several smaller files keep the template organized:

- `student-information.tex` stores the student name, dissertation title, degree program, degree year, committee information, and PDF metadata text.
- `front-accessible.tex` stores the front matter, including the abstract, dedication, biography, acknowledgements, table of contents, list of tables, list of figures, and authorship statement.
- `bibliography.bib` stores the bibliography entries.
- `ncsu-required.tex` stores required NC State formatting and accessibility settings.
- `student-macros.tex` stores student-specific packages, commands, notation, and macros.
- `optional-accessible.tex` stores optional visual choices, such as font choices, chapter heading style, and theorem box style.

The main chapters are stored in separate folders, such as `Chapter-1/Chapter-1.tex`, `Chapter-2/Chapter-2.tex`, and `Chapter-3/Chapter-3.tex`. Appendices are stored in separate appendix folders. Keeping this folder structure intact is important. If you rename a file or folder, you must also update the corresponding `\include{...}` line in the main file.

This template must be compiled with LuaLaTeX. The sample chapters are placeholder material only. They are included to test chapter structure, figures, tables, mathematical content, citations, appendices, and accessibility features. Much of the placeholder content was generated with AI.

Title of This Dissertation

by
Firstname Middlename Lastname

A dissertation submitted to the Graduate Faculty of
North Carolina State University
in partial fulfillment of the
requirements for the Degree of
Doctor of Philosophy

Department of Whichever

Raleigh, North Carolina
2026

APPROVED BY:

Member 1	Member 2
Member 3	Member 4

Chair Person
Chair of Advisory Committee

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When you begin adapting the template, start with small edits. First, open `student-information.tex` and update the student name, dissertation title, committee information, program name, degree year, PDF subject, and PDF keywords. Then compile. Important reminder: this template must be compiled with LuaLaTeX.

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- Write your main thesis or dissertation content in the chapter files, such as `Chapter-1/Chapter-1.tex`, `Chapter-2/Chapter-2.tex`, and so on.

- Write your abstract, dedication, biography, acknowledgements, and authorship statement in `front-accessible.tex`.

- Write appendix material in the appendix files, such as `Appendix-A/Appendix-A.tex` and `Appendix-B/Appendix-B.tex`. Appendices are a good place for supporting material, longer tables, extra figures, code documentation, survey instruments, or proofs that do not belong in the main chapters.

- Put cited works in `bibliography.bib`. Each source should have a BibTeX entry with a citation key, and your chapters should cite those sources using the matching key.

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The file `ncsu-required.tex` contains required template settings for the accessible NC State thesis or dissertation build. The file `optional-accessible.tex` contains optional visual choices. If something is only a design preference, it belongs in `optional-accessible.tex`, not in `ncsu-required.tex`. Options in `optional-accessible.tex` include font choices and theorem environments.

BIOGRAPHY

This Biography is being used for accessibility guidance. Before final submission, replace this text with your actual biography.

The goal of this template is to make the final PDF more accessible than a traditional LaTeX thesis or dissertation PDF. The template can help with tagging, structure, metadata, figure descriptions, and mathematical accessibility, but accessibility also depends on the choices you make while writing.

Use real LaTeX structure. Write chapters with `\chapter`, sections with `\section`, and subsections with `\subsection`. Do not simulate headings by manually changing font size or using bold text alone.

Write equations in LaTeX. Do not insert equations as screenshots. Use labels for equations, figures, tables, sections, and chapters that you need to refer to later. Do not manually type equation numbers, figure numbers, or table numbers.

Give every meaningful figure a real caption and meaningful alt text. You do not need to rewrite every figure using a special command. The usual LaTeX `figure` environment is fine. The important accessibility step is to include meaningful alt text in the `\includegraphics` command.

For example:

```
\begin{figure}[htbp]
  \centering
  \includegraphics[
    width=\textwidth,
    alt={Briefly describe the important visual content of the figure.}
  ]{figures/example.png}
  \caption{Caption explaining how the figure is used in the dissertation.}
  \label{fig:example}
\end{figure}
```

If you already have many figures, you usually only need to add or improve the `alt={...}` text inside each `\includegraphics` command. You do not need to change the whole structure of every figure.

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For mathematical figures, describe the mathematical structure, not every decorative detail. Do not rely only on color to communicate information.

Use real table structures for tables, and make sure tables have clear column headers. Do not use screenshots of tables unless there is no reasonable alternative. Try to keep tables simple, because complicated tables are harder to read visually and harder to make accessible. Tables must have captions.

If a figure caption contains mathematics, use a math-free optional short caption so the List of Figures does not contain tagged formula content. For example, use

```
\caption[The Pythagorean Theorem]{The Pythagorean Theorem says \(a^2 + b^2 = c^2.\
```

This allows the full caption under the figure to contain mathematical notation while keeping the List of Figures easier for validation tools and assistive technologies to process. You will get errors in VeraPDF if you do not use the above structure, even if the document compiles and looks fine to you.

The same idea applies to section headings and chapter headings. If a heading contains substantial mathematics, Unicode symbols, or unusual characters, consider using a simpler text version in the optional argument. Headings and captions are reused in bookmarks, the table of contents, the list of figures, and the list of tables, so simple text versions are often more reliable.

Automated accessibility checkers are useful, but they do not catch everything. A PDF can pass an automated check and still be hard to use, so manual review still matters.

ACKNOWLEDGEMENTS

These Acknowledgements are being used for compiling and validation instructions. Before final submission, replace this text with your actual acknowledgements.

- In TeXShop or another local editor, open `dissertation-main-accessible.tex` and choose LuaLaTeX before compiling. A full dissertation usually needs more than one compile run. This is normal. LaTeX needs extra runs to build the table of contents, citations, cross-references, list of figures, and list of tables.
- In Overleaf, set the compiler to LuaLaTeX and set the main document to `dissertation-main-accessible.tex`. The safest way to begin in Overleaf is to upload the whole project as a zip file. Uploading files one at a time can accidentally flatten the folder structure, which may cause errors when LaTeX tries to find chapter files.
- Use `student-information.tex` for the student name, dissertation title, degree program, degree year, committee information, and PDF metadata.
- Use `student-macros.tex` for student-specific packages, notation, and shortcuts. This is the right place for commands that are specific to your dissertation.
- Leave `ncsu-required.tex` in place for the final accessible NC State build. This file contains required formatting, hyperlink, PDF metadata, bookmark, caption, and theorem-like environment setup.
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```
LuaLaTeX
BibTeX
LuaLaTeX
LuaLaTeX
```

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If references, citations, figure numbers, table numbers, or the table of contents look wrong, compile again before assuming something is broken. If they still look wrong after several runs, check for misspelled labels or citation keys. The repeated LuaLaTeX runs are what allow LaTeX to update the table of contents, list of figures, list of tables, citations, and cross-references.

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```
This is a VeraPDF validation report for my LaTeX dissertation PDF.
Please explain what the errors mean, which ones are likely caused by
my LaTeX source, and what files or commands I should check first.
```

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The goal is to support more accessible thesis and dissertation PDFs and to aim for accessible, reusable structure wherever possible. Using this template does not guarantee compliance.

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AUTHORSHIP STATEMENT

List contributions clearly by chapter. Use normal list structure rather than manually bolded paragraphs. Replace this sample authorship statement with the statement required for your dissertation, if applicable.

Chapter 1. Student Name: sole author of Chapter 1.

Chapter 2. Student Name: primary writing, coding, analysis, and literature review. Advisor Name: project guidance, editing, and revision.

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CHAPTER 1

The Fundamental Theorem of Line Integrals

1.1 Purpose of This Sample Chapter

This chapter is included to show that the template can support ordinary dissertation writing, mathematical exposition, displayed equations, theorem environments, proofs, diagrams, tables, citations, and figures with alt text. A student using this file should replace the mathematical content with their own work, but the structure is intended to model accessible source habits.

In particular, this chapter demonstrates the following practices:

1. Use real section and subsection headings instead of visual formatting alone.
2. Write equations in LaTeX source, not as images.
3. Give every meaningful figure alternative text in the source file.
4. Use tables with header rows, not aligned text made with spaces.
5. Use descriptive captions that explain why the figure or table appears in the chapter.

1.2 A First Definition

Line integrals give a way to accumulate information along a curve. In vector calculus, one common version measures the component of a vector field in the direction of motion along a parametrized path.

Definition 1.1: Line integral of a vector field. Let C be a smooth curve parametrized by $\mathbf{r}(t)$, where $a \le t \le b$. Let $\mathbf{F}$ be a vector field defined on the curve. The line integral of $\mathbf{F}$ over C is

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt.$$

The notation $d\mathbf{r}$ reminds us that the integral follows the direction of the curve. If the parametrization runs from a starting point P to an ending point Q, then the sign and value of the integral can depend on that direction.

1.3 The Fundamental Theorem

The fundamental theorem of line integrals is the vector-calculus analogue of the one-variable fundamental theorem of calculus. It says that if a vector field is the gradient of a scalar function, then the line integral depends only on the endpoint values of that scalar function.

Below, we see two subsections.

1.3.1 Statement of the theorem

Theorem 1.2: Fundamental theorem of line integrals. *Let f be a differentiable scalar-valued function whose gradient is continuous on an open region containing a smooth curve C. Suppose C is parametrized by $\mathbf{r}(t)$, where $a \le t \le b$. Then*

$$\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a)).$$

This result is powerful because it replaces an integral over an entire curve with a calculation at two points. When $\mathbf{F} = \nabla f$, the function f is called a potential function for $\mathbf{F}$, and the vector field is called conservative.

1.3.2 Proof of the theorem

Proof. Since C is parametrized by $\mathbf{r}(t)$, Definition 1.1 gives

$$\int_C \nabla f \cdot d\mathbf{r} = \int_a^b \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt.$$

By the multivariable chain rule,

$$\frac{d}{dt} f(\mathbf{r}(t)) = \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t).$$

Therefore,

$$\int_C \nabla f \cdot d\mathbf{r} = \int_a^b \frac{d}{dt} f(\mathbf{r}(t))\, dt = f(\mathbf{r}(b)) - f(\mathbf{r}(a)),$$

by the one-variable fundamental theorem of calculus. $\square$

1.4 A Diagram of the Idea

Figure 1.1 shows a curve that begins at P and ends at Q. The theorem says that when $\mathbf{F} = \nabla f$, the integral along the curve is determined by the value of f at those two endpoints, not by the particular route between them.

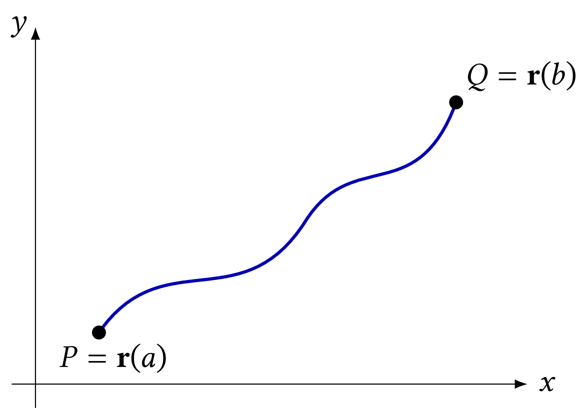

Figure 1.1: A path from P to Q. For a conservative vector field, the line integral depends only on value of the potential function at the endpoints.

1.5 A Short Computation

The theorem can be checked in a concrete case. Let

$$f(x, y) = x^2 + y^2,$$

so that

$$\nabla f(x, y) = \langle 2x, 2y \rangle.$$

Let C be parametrized by $\mathbf{r}(t) = \langle t, t^2 \rangle$, where $0 \le t \le 1$. Then $\mathbf{r}(0) = \langle 0, 0 \rangle$ and $\mathbf{r}(1) = \langle 1, 1 \rangle$. By Theorem 1.2,

$$\int_C \nabla f \cdot d\mathbf{r} = f(1,1) - f(0,0) = 2.$$

We can also compute directly from the parametrization:

$$\begin{aligned}
\int_C \nabla f \cdot d\mathbf{r} &= \int_0^1 \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt \\
&= \int_0^1 \langle 2t, 2t^2 \rangle \cdot \langle 1, 2t \rangle\, dt \\
&= \int_0^1 \left(2t + 4t^3\right)\, dt \\
&= \left[t^2 + t^4\right]_0^1 \\
&= 2.
\end{aligned}$$

This agrees with the endpoint calculation.

1.6 A Small Table Example

Tables should be used for data that are genuinely tabular. Table 1.1 compares the two approaches used in the example above.

Table 1.1: Two ways to compute the same conservative line integral.

Method	**Main calculation**	**Value**
Endpoint method	Compute $f(\mathbf{r}(1)) - f(\mathbf{r}(0))$.	2
Direct integral	Compute $\int_0^1 \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt$.	2

1.7 An Example with Citations

A dissertation chapter usually includes citations as part of the mathematical or scientific narrative. Parenthetical citations can be written with `\citep`, as in (**?**). Narrative citations can be written with `\citet`, as in **?**. Students should replace these sample references with sources that are relevant to their own work.

1.8 A fancy diagram

Many dissertations in mathematics use diagrams to explain structure. In algebraic topology, algebraic geometry, and homological algebra, commutative diagrams are often used to organize maps between groups, modules, or vector spaces. This chapter gives a sample of how a mathematical dissertation might combine definitions, a theorem, a proof sketch, and an accessible TikZ figure.

The example is the snake lemma. The name comes from the connecting homomorphism that winds through the diagram from a kernel term to a cokernel term.

Commutative diagrams are commonly used to organize algebraic information in topology and homological algebra (Rivera and Chen 2021). The snake lemma is a standard example of a result whose proof is naturally represented by a diagram (Patel 2020).

1.9 Exact Sequences

A sequence of groups and homomorphisms

$$A \xrightarrow{f} B \xrightarrow{g} C \tag{1.1}$$

is said to be exact at B if

$$\mathrm{im}(f) = \ker(g). \tag{1.2}$$

In words, exactness says that the elements of B that are sent to zero by g are exactly the elements that came from A.

A short exact sequence has the form

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0. \tag{1.3}$$

This means that f is injective, g is surjective, and the image of f is the kernel of g.

1.10 The Snake Lemma

The snake lemma starts with a commutative diagram whose rows are exact:

$$0 \longrightarrow A' \longrightarrow B' \longrightarrow C' \tag{1.4}$$

and

$$0 \longrightarrow A \longrightarrow B \longrightarrow C. \tag{1.5}$$

There are also vertical maps from the top row to the bottom row. The conclusion of the snake lemma is that these maps produce a new exact sequence involving kernels and cokernels.
Theorem: Snake Lemma. Suppose the rows in Figure 1.2 are exact and the diagram commutes. Then there is a connecting homomorphism

$$\delta : \ker(\gamma) \longrightarrow \operatorname{coker}(\alpha) \tag{1.6}$$

such that the following sequence is exact:

$$\ker(\alpha) \longrightarrow \ker(\beta) \longrightarrow \ker(\gamma) \xrightarrow{\delta} \operatorname{coker}(\alpha) \longrightarrow \operatorname{coker}(\beta) \longrightarrow \operatorname{coker}(\gamma). \tag{1.7}$$

1.11 The Diagram

Figure 1.2 shows the standard snake lemma diagram. The solid arrows show the original commutative diagram. The dashed curved arrow shows the connecting homomorphism, which is the part of the diagram that gives the lemma its name.

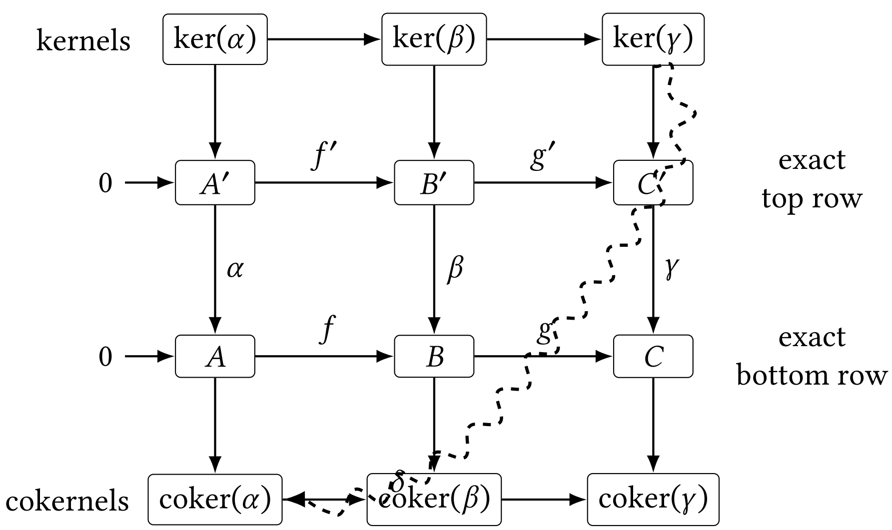

Figure 1.2: A schematic snake lemma diagram. The solid arrows show the original commutative diagram with exact rows. The dashed winding arrow represents the connecting homomorphism from $\ker(\gamma)$ to $\operatorname{coker}(\alpha)$.

1.12 Constructing the Connecting Homomorphism

The connecting homomorphism is the key part of the theorem. The construction follows one element through the diagram.

Let $c' \in \ker(\gamma)$. Since $c' \in C'$ and the top row is exact, we choose an element $b' \in B'$ such that

$$g'(b') = c'. \tag{1.8}$$

Now apply the vertical map β. Since the diagram commutes,

$$g(\beta(b')) = \gamma(g'(b')) = \gamma(c') = 0. \tag{1.9}$$

Thus $\beta(b') \in \ker(g)$. Since the bottom row is exact, there is an element $a \in A$ such that

$$f(a) = \beta(b'). \tag{1.10}$$

The image of a in $\operatorname{coker}(\alpha)$ is defined to be $\delta(c')$. In other words,

$$\delta(c') = [a] \in \operatorname{coker}(\alpha). \tag{1.11}$$

The proof then checks that this definition does not depend on the choices made, that δ is a homomorphism, and that the resulting sequence is exact.

1.13 Why the Diagram Matters

The snake lemma is useful because it turns a large commutative diagram into an exact sequence. In algebraic topology, this kind of result helps compare homology groups that arise from related spaces, pairs of spaces, or chain complexes.

The diagram is not just decorative. It records the logic of the proof:

1. Start with an element in $\ker(\gamma)$.
2. Lift the element across the top row.
3. Move down through the vertical map.
4. Use exactness of the bottom row.
5. Record the resulting class in $\operatorname{coker}(\alpha)$.

The winding arrow in the figure summarizes this path through the diagram.

1.14 Accessibility Notes for Mathematical Diagrams

Commutative diagrams can be difficult to interpret with assistive technology. A thesis should not rely on the image alone. The surrounding text should describe the structure of the diagram and explain how the diagram is used in the proof.

For this kind of figure, the alt text should identify the major objects, the direction of the maps, and the role of the special arrow. The caption should explain what the figure represents in the argument. The body text should then walk through the important path in the diagram.

1.15 Summary

This chapter illustrates how a mathematical dissertation chapter might combine exposition, definitions, theorems, proofs, diagrams, computations, tables, citations, and accessible figures. The mathematical content is an AI-generated sample. The important template lesson is that accessibility starts in the source: headings, equations, captions, alt text, tables, and links should be written carefully before the final PDF is created.

CHAPTER 2

Synthesis and Characterization of a Model Organic Compound

2.1 Purpose of This Sample Chapter

This chapter is written as a short example of how a dissertation chapter in organic chemistry might organize a synthesis, reaction optimization, and characterization discussion in an accessible LaTeX document. Here are two sample citations (Morrison 2019) and (Okafor and Hughes 2022).

Organic chemistry chapters often move between narrative explanation, reaction schemes, tabular summaries, and compact characterization data. The accessibility goal is the same as in any other discipline: do not rely on a picture alone. A reaction scheme should have a caption and alt text, a table should use real rows and columns, and important observations from a spectrum should be described in nearby text.

2.2 Reaction Design

As a model example, consider the reduction of acetophenone to 1-phenylethanol. The reaction is common enough to be recognizable, but simple enough to be used in a template without requiring a specialized chemistry drawing package. Figure 2.1 gives a simplified reaction scheme.

The reaction is an example of nucleophilic hydride delivery to a ketone. In words, a hydride equivalent adds to the carbonyl carbon, and protonation during workup gives the corresponding secondary alcohol. If the stereochemical outcome matters for the research question, then the chapter should say whether the product is racemic, enantioenriched, or isolated as a single stereoisomer. In this short example, the product is treated as racemic.

$$\begin{array}{c}\mathrm{PhCOCH_3}\\ \text{acetophenone}\\ \mathbf{1}\end{array} \xrightarrow[\text{0°C to room temperature; aqueous workup}]{\mathrm{NaBH_4},\ \mathrm{MeOH}} \begin{array}{c}\mathrm{PhCH(OH)CH_3}\\ \text{1-phenylethanol}\\ \mathbf{2}\end{array}$$

Figure 2.1: A simplified reaction scheme for reducing acetophenone to 1-phenylethanol. A student preparing a real organic chemistry dissertation would usually replace this text-based scheme with a properly drawn chemical structure and an equally descriptive alternative text.

2.3 General Experimental Approach

A typical experimental section gives enough information for a trained chemist to understand what was done and, when appropriate, repeat the experiment. It usually includes the scale, solvent, temperature, atmosphere, purification method, yield, and characterization data. The prose should be compact, but not cryptic.

For example, the following paragraph gives a template-style description of the model reaction. The quantities are included only as examples and should be checked before being used in any actual experiment.

> A flame-dried round-bottom flask equipped with a magnetic stir bar was charged with acetophenone **1** (1.00 mmol) and methanol (5 mL). The solution was cooled to 0°C, and sodium borohydride (1.20 mmol) was added portionwise. The reaction mixture was allowed to warm to room temperature and was stirred until thin-layer chromatography indicated consumption of the starting material. The reaction was quenched with saturated aqueous ammonium chloride, extracted with ethyl acetate, dried over sodium sulfate, filtered, and concentrated under reduced pressure. Purification by flash column chromatography gave 1-phenylethanol **2** as a colorless oil.

This kind of paragraph contains several accessibility concerns. Abbreviations such as TLC should be spelled out on first use. Chemical hazards and specialized procedures should not be hidden in a figure. If color changes, precipitate formation, or chromatographic behavior are important, they should be described in text rather than only shown in a photograph.

2.4 Optimization Data

Reaction optimization is often summarized in a table. Table 2.1 gives a small example. The table uses real text, real column headers, and a visible caption. It is not a screenshot of a spreadsheet.

Table 2.1: Example optimization results for the model reduction of acetophenone. The data are illustrative and are included to show accessible table structure.

Entry	Hydride source	Solvent	Temperature	Isolated yield
1	$NaBH_4$	Methanol	Room temperature	82%
2	$NaBH_4$	Ethanol	Room temperature	76%
3	$NaBH_4$	Methanol	0°C to room temperature	88%
4	$LiAlH_4$	Diethyl ether	0°C	71%

In the narrative following a table, describe the main pattern rather than forcing readers to infer the conclusion from the table alone. In this example, entry 3 gives the highest isolated yield under the tested conditions. The lower yield in ethanol may reflect slower reduction, while the use of lithium aluminum hydride gives no advantage for this simple substrate.

2.5 Characterization Data

Characterization data are often compact, but they should still be readable. A student should define unfamiliar abbreviations and explain which data support the structural assignment. For example, proton nuclear magnetic resonance spectroscopy, often abbreviated ^{1}H NMR, can distinguish the methyl group and the benzylic proton in 1-phenylethanol. Carbon nuclear magnetic resonance spectroscopy, or ^{13}C NMR, gives complementary evidence for the aromatic carbons and the alcohol-bearing carbon.

Table 2.2: Example characterization summary for 1-phenylethanol. The values are illustrative placeholders for a template and should be replaced with verified experimental values.

Technique	Example observation
^{1}H NMR	Aromatic protons appear near δ7.2–7.4. The benzylic proton appears downfield relative to the methyl group.
^{13}C NMR	Aromatic carbon signals and an oxygen-bearing benzylic carbon are observed.
Infrared spectroscopy	A broad absorption consistent with an alcohol $O-H$ stretch is observed.
High-resolution mass spectrometry	The measured mass should be reported with the calculated mass for the proposed molecular formula.

A short characterization paragraph might read as follows:

> Compound **2** was obtained as a colorless oil. The ^{1}H NMR spectrum showed a set of aromatic resonances and signals consistent with a benzylic methine proton and a methyl group. The infrared spectrum contained a broad absorption consistent with an alcohol functional group. These data support assignment of the reduced alcohol product rather than the starting ketone.

For a real dissertation, the student should replace this paragraph with the full characterization format expected by the discipline, research group, and journal style. The key accessibility point is that the text should state the conclusion supported by the data. A spectrum image may be useful, but it should not be the only place where the structural evidence is communicated.

2.6 A Small Stoichiometric Calculation

Organic chemistry chapters sometimes include short calculations, especially when discussing scale, equivalents, yield, or selectivity. For example, the percent yield is computed from the isolated amount and the theoretical amount:

$$\text{percent yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \cdot 100\%. \tag{2.1}$$

If the theoretical yield is 122.17 mg and the isolated product is 107.5 mg, then

$$\text{percent yield} = \frac{107.5\,\text{mg}}{122.17\,\text{mg}} \cdot 100\% \tag{2.2}$$

$$= 88.0\%. \tag{2.3}$$

This calculation is included as text-based mathematics rather than as an image, so it can be enlarged, searched, copied, and interpreted by assistive technology more reliably than a screenshot.

2.7 Accessibility Notes for Chemistry Content

Chemistry documents can be difficult to make accessible because important meaning is often packed into visual structures, spectra, and schemes. The following practices help:

1. Give each reaction scheme a caption that states the purpose of the scheme.

2. Provide alt text that identifies the starting material, product, reagents, and main transformation.

3. Summarize the important conclusion from each spectrum in text.
4. Use real tables for optimization and screening data.
5. Avoid using a screenshot when a table, equation, or short text description would work.

These habits do not replace discipline-specific expectations. They make those expectations easier to meet in a way that is more usable for readers with different access needs.

CHAPTER 3

The Central Limit Theorem

The central limit theorem is one of the main reasons that the normal distribution appears throughout statistics. Roughly speaking, it says that averages of many independent observations tend to look normal, even when the original data do not come from a normal distribution.

This AI-generated chapter gives a short example of how a statistics dissertation might present definitions, a theorem, a proof sketch, a table, and a figure. The goal is model accessible writing in a statistical context.

3.1 Sample Means

Suppose that $X_1, X_2, \dots, X_n$ are independent observations from the same population. The sample mean is

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i. \tag{3.1}$$

The sample mean is often used to estimate the population mean. If the population mean is μ, then $\bar{X}_n$ tends to get closer to μ as the sample size increases. This is the content of the law of large numbers.

The central limit theorem gives more information. It describes the shape of the distribution of the sample mean after it has been centered and scaled.

3.2 Statement of the Theorem

Theorem: Central Limit Theorem. Suppose that $X_1, X_2, \dots, X_n$ are independent and identically distributed random variables with mean μ and finite, positive variance σ^2. Then

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} Z, \tag{3.2}$$

where Z has the standard normal distribution.

In words, the standardized sample mean becomes approximately standard normal as n becomes large. The notation $\xrightarrow{d}$ means convergence in distribution.

3.3 Interpretation

The expression in Equation 3.2 has three important pieces.

1. The numerator $\bar{X}_n - \mu$ centers the sample mean at zero.
2. The denominator $\sigma/\sqrt{n}$ rescales the sample mean by its standard deviation.
3. The conclusion says that this centered and scaled quantity has an approximately normal distribution for large n.

This result is useful because it allows researchers to use normal approximations in many settings where the original data are not normally distributed.

For example, suppose that the original observations are right-skewed. Individual observations may not look symmetric, but the averages of many such observations may be much closer to normal.

3.4 A Proof Sketch

A full proof of the central limit theorem requires more probability theory than we want to include in this sample chapter. However, the main idea can be explained using moment generating functions.

Let

$$Y_i = \frac{X_i - \mu}{\sigma}. \tag{3.3}$$

Then each Y_i has mean 0 and variance 1. The standardized sample mean can be written as

$$
\begin{aligned}
\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} &= \frac{\sqrt{n}}{\sigma}\left(\bar{X}_n - \mu\right) && (3.4)\\
&= \frac{\sqrt{n}}{\sigma}\left(\frac{1}{n}\sum_{i=1}^{n} X_i - \mu\right) && (3.5)\\
&= \frac{1}{\sqrt{n}}\sum_{i=1}^{n} \frac{X_i - \mu}{\sigma} && (3.6)\\
&= \frac{1}{\sqrt{n}}\sum_{i=1}^{n} Y_i. && (3.7)
\end{aligned}
$$

The proof then studies the distribution of the sum in Equation 3.7. Under the hypotheses of the theorem, the moment generating function of this standardized sum approaches the moment generating function of a standard normal random variable. Since moment generating functions determine distributions under appropriate conditions, the standardized sample mean converges in distribution to a standard normal random variable.

3.5 A Simulation Example

Table 3.1 shows a schematic example of what one might observe in a simulation. The original data are assumed to come from a skewed population. As the sample size increases, the distribution of the sample mean becomes more symmetric and more concentrated around the population mean.

Table 3.1: A schematic summary of how sample means behave as the sample size increases.

Sample size	Shape of sample mean distribution	Variability
$n = 5$	Still noticeably skewed	Relatively large
$n = 30$	More symmetric	Smaller
$n = 100$	Close to normal in many cases	Much smaller

The table is not meant to replace a formal theorem. Instead, it gives a practical interpretation of the theorem: larger samples tend to produce sample means whose distributions are more normal and less spread out.

3.6 A Diagram of the Normal Approximation

Figure 3.1 shows the standard normal curve that appears in the conclusion of the central limit theorem. In an accessible document, the caption identifies the figure, and the surrounding text explains why it matters.

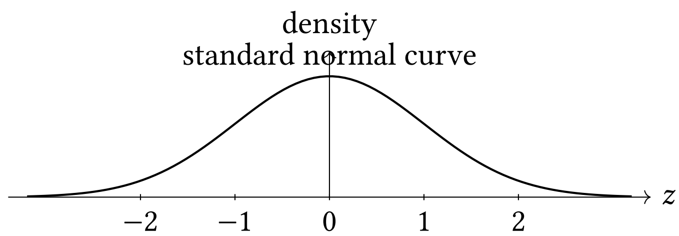

Figure 3.1: The standard normal curve used in the central limit theorem. After centering and scaling, the sample mean is approximately distributed according to this curve when the sample size is large.

3.7 Why This Matters

The central limit theorem supports many common statistical procedures. For example, confidence intervals and hypothesis tests for means often rely on the fact that standardized sample means are approximately normal.

A typical confidence interval for a population mean has the form

$$\bar{X}_n \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \tag{3.8}$$

when σ is known and the sample size is large. Here, $z_{\alpha/2}$ is chosen from the standard normal distribution.

In practice, σ is often unknown and must be estimated from the data. That leads to related procedures involving the sample standard deviation and the t-distribution. The central idea is the same: the sample mean is random, but its distribution can often be approximated well enough to make statistical inference possible.

3.8 Examples of Image Inclusion

The following figure is intentionally left from the original sample file. It is included here to demonstrate the accessibility helper commands for ordinary image files and cropped image files.

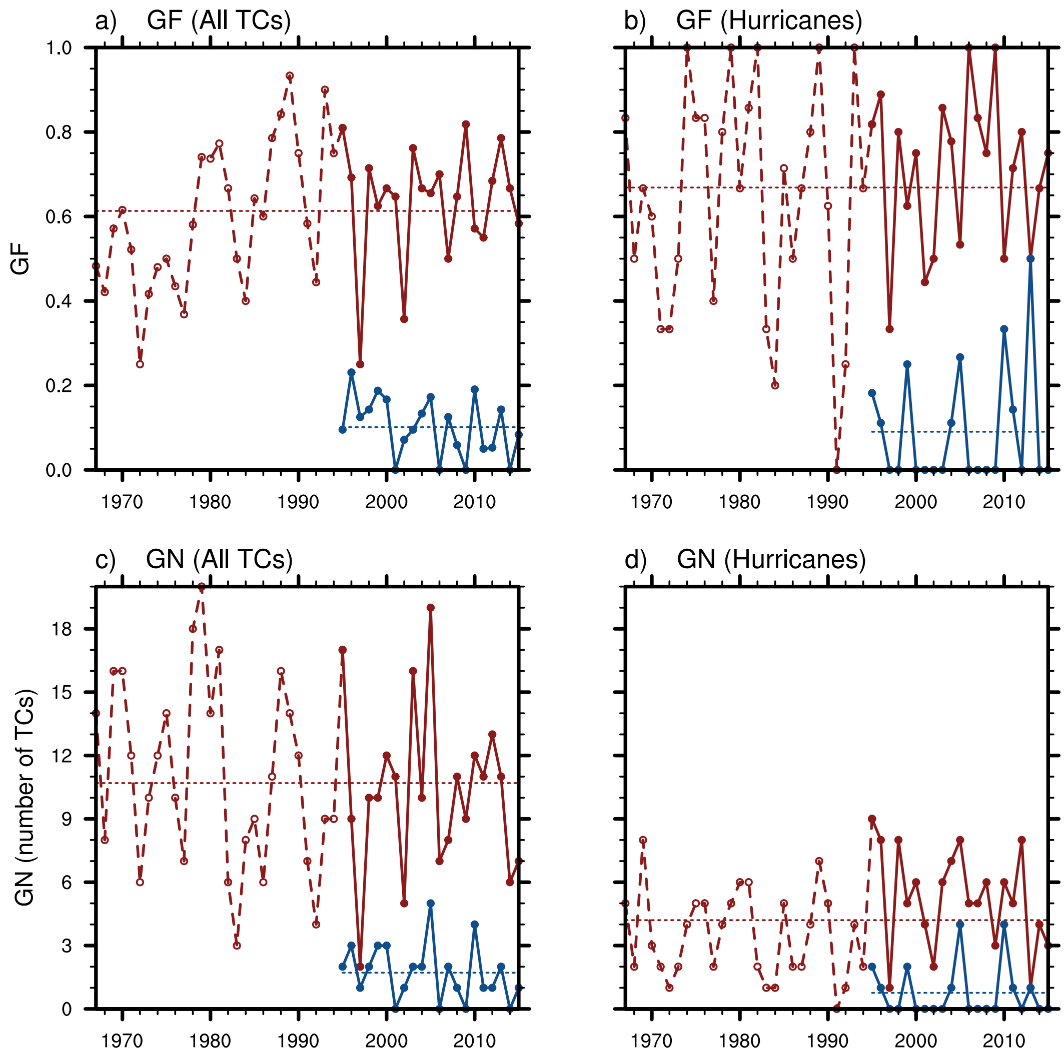

Figure 3.2: Sample cropped image included with the original template. Replace this image, alt text, and caption with content from the dissertation.

3.9 Accessibility Notes for Statistical Writing

Statistical writing may include formulas, tables, plots, and notation-heavy explanations. To make this material more accessible, authors should follow these practices.

1. Explain the meaning of each major equation in nearby text.
2. Use real LaTeX equations rather than screenshots of equations.
3. Use real tables with column headers rather than images of tables.
4. Describe the purpose and main message of each plot in the surrounding text.
5. Avoid relying on color alone to communicate statistical conclusions.

These practices help readers who use screen readers, readers who enlarge text, and readers who are new to the notation.

CHAPTER 4

Marine, Earth, and Atmospheric Science

Marine, earth, and atmospheric science often connects observations from the ocean, atmosphere, and land surface. A dissertation in this area may include maps, satellite images, time series, model output, field measurements, and physical equations. This sample chapter shows how that kind of material can be presented in an accessible way.

The example focuses on tropical weather systems. The goal is not to give a complete scientific treatment, but to demonstrate how a student might combine text, figures, tables, and equations in a thesis or dissertation.

4.1 Scientific Context

Tropical cyclones are influenced by several environmental conditions, including sea surface temperature, atmospheric moisture, vertical wind shear, and large-scale circulation patterns. These variables interact across different spatial scales. For example, warm ocean water can provide energy for convection, while strong vertical wind shear can disrupt storm organization.

Figure 4.1 shows an example of a weather-related image from the original template. In a real dissertation, the caption should explain the role of the figure in the argument, and the alt text should describe the visual information needed by a reader who cannot see the image.

4.2 Environmental Variables

A typical analysis may compare several environmental variables before, during, and after a storm event. Table 4.1 gives a simple example. The table uses real text and column headers rather than a screenshot.

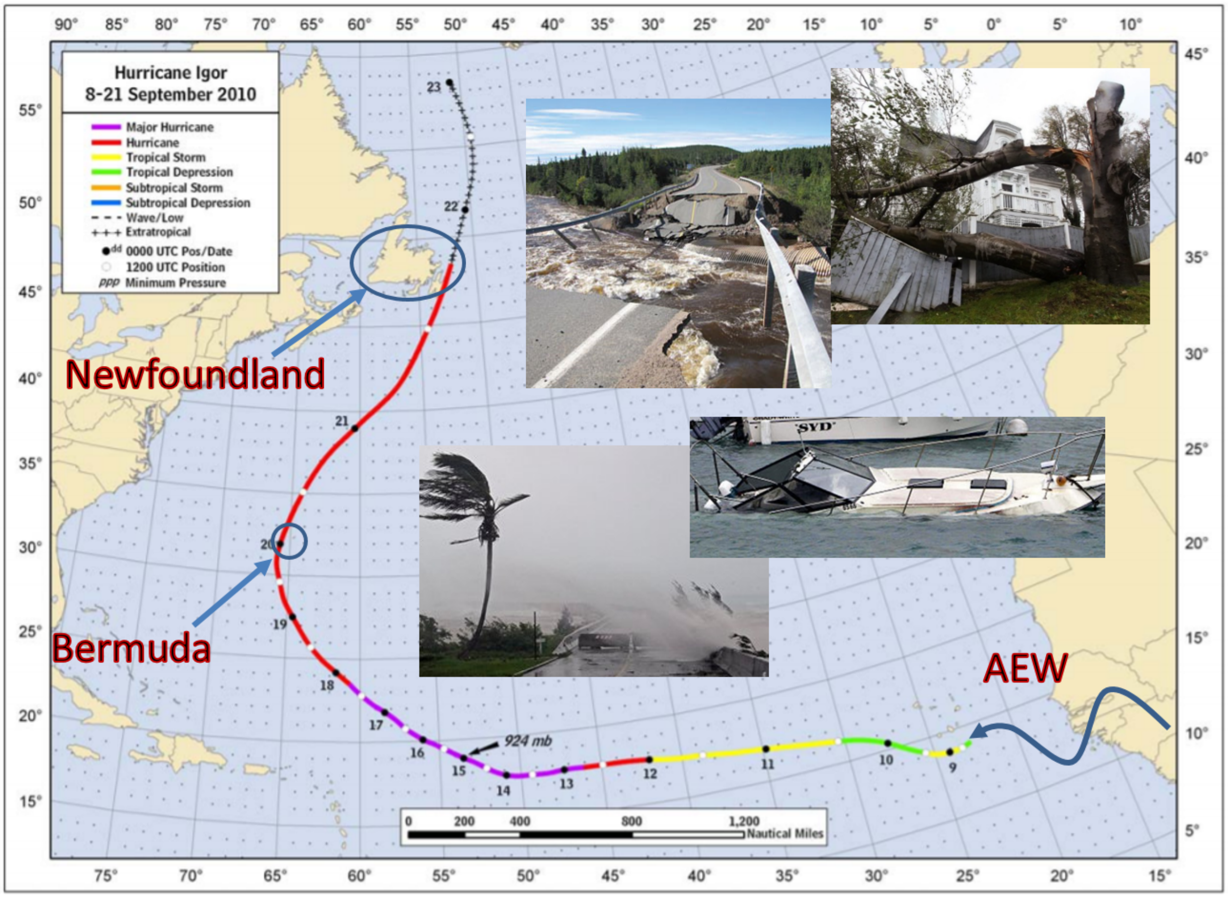

Figure 4.1: Sample weather-related image included with the original template. In a real dissertation, replace this image, alt text, and caption with content from the dissertation.

Table 4.1: Examples of environmental variables used in tropical weather analysis.

Variable	Common interpretation	Example unit
Sea surface temperature	Ocean energy available to the atmosphere	˚C
Vertical wind shear	Change in wind with height	m/s
Relative humidity	Atmospheric moisture content	percent
Surface pressure	Strength of low-pressure circulation	hPa
Precipitation rate	Rainfall intensity	mm/hr

Tables like this are useful for readers because they define variables before the variables appear in formulas or figures.

4.3 A Basic Physical Relationship

One common approximation in atmospheric science is hydrostatic balance. It relates the vertical pressure gradient to the density of air and the acceleration due to gravity:

$$\frac{\partial p}{\partial z} = -\rho g. \tag{4.1}$$

In Equation 4.1, p is pressure, z is height, ρ is air density, and g is the acceleration due to gravity. The negative sign indicates that pressure decreases as height increases.

This equation is not a complete model of the atmosphere. Instead, it is a useful approximation when vertical accelerations are small compared with the force of gravity.

4.4 A Schematic Cross Section

Figure 4.2 gives a simple schematic of a tropical system over warm ocean water. The figure is drawn with TikZ so that the source remains editable. The alt text describes the main visual features of the diagram.

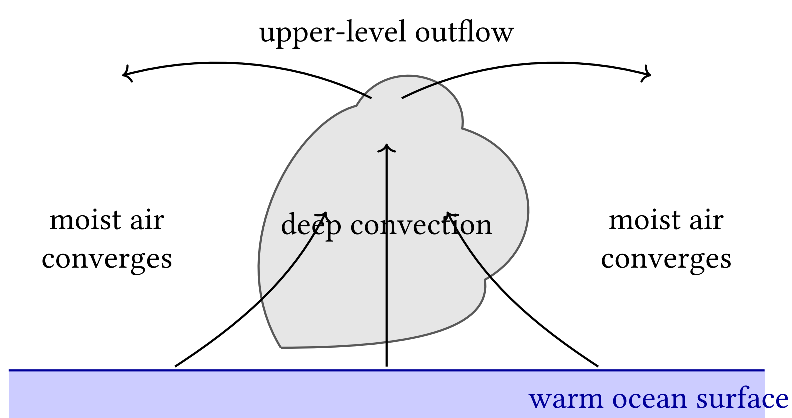

Figure 4.2: A schematic cross section of a tropical weather system. Warm ocean water supports rising moist air, deep convection, and upper-level outflow.

4.5 Example Analysis

Suppose a researcher wants to compare the environment before and during a storm event. One simple approach is to compute changes in variables over time. For example, if T_0 is the sea surface temperature before the event and T_1 is the sea surface temperature during the event, then the temperature change is

$$\Delta T = T_1 - T_0. \tag{4.2}$$

A negative value of ΔT indicates cooling, while a positive value indicates warming. In a real analysis, this calculation might be repeated across many grid points, buoys, or satellite retrievals.

For a spatially averaged value over a region R, one might write

$$\overline{T} = \frac{1}{A(R)} \int_R T(x, y)\, dA, \tag{4.3}$$

where $A(R)$ is the area of the region and $T(x, y)$ is the temperature at location (x, y). The surrounding text should explain the region, the data source, and the units.

4.6 Accessibility Notes for Earth Science Figures

Earth science dissertations often rely on highly visual material. Maps, radar images, satellite images, and model-output plots can be difficult to interpret without careful explanation. Authors should follow these practices.

1. Describe the main visual message of each map or plot in nearby text.
2. Include alt text for every meaningful figure.
3. Avoid relying on color alone to distinguish categories or intensities.
4. Use readable labels, legends, and units.
5. Do not use screenshots of tables or equations.
6. Explain abbreviations such as SST, RH, MSLP, or AEW the first time they appear.

These practices help readers understand the scientific argument even when they cannot rely on the visual display alone.

CHAPTER 5

Computational Methods in Engineering

Many dissertations in computer science, engineering, applied mathematics, and the physical sciences include algorithms, source code, numerical experiments, or computational workflows. This sample chapter shows one way to present code in a thesis while keeping the material readable and accessible.

The main example is a finite difference method for the one-dimensional heat equation. The goal is not to provide a complete numerical analysis, but to demonstrate how a student might present a model problem, an algorithm, a short Python script, a short MATLAB script, and a table of computational results.

5.1 A Model Problem

Consider the one-dimensional heat equation

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}, \tag{5.1}$$

where $u(x,t)$ is the temperature at position x and time t, and α is the thermal diffusivity.

For this demonstration, suppose the spatial domain is $0 \leq x \leq 1$, and suppose the boundary temperatures are fixed:

$$u(0,t) = 0 \qquad \text{and} \qquad u(1,t) = 0. \tag{5.2}$$

We also choose the initial temperature distribution

$$u(x,0) = \sin(\pi x). \tag{5.3}$$

This problem is simple enough to explain in a template, but it still looks like the kind of model problem that appears in engineering and scientific computing.

5.2 Finite Difference Approximation

A finite difference method replaces derivatives with difference quotients. If $x_j = j\Delta x$ and $t_n = n\Delta t$, then we write u_j^n for an approximation to $u(x_j, t_n)$.

A common explicit approximation is

$$u_j^{n+1} = u_j^n + r\left(u_{j-1}^n - 2u_j^n + u_{j+1}^n\right), \tag{5.4}$$

where

$$r = \frac{\alpha\Delta t}{(\Delta x)^2}. \tag{5.5}$$

The value r is often called the stability ratio. For the explicit method, the time step must be chosen carefully. If r is too large, the numerical solution may become unstable.

5.3 Algorithm

The method can be summarized as follows.

1. Choose the number of spatial grid points and the final time.
2. Compute Δx, Δt, and r.
3. Set the initial condition at each grid point.
4. Apply the boundary conditions.
5. Step forward in time using Equation 5.4.
6. Store or plot the approximate solution.

When writing for accessibility, it is helpful to describe the algorithm in prose before showing code. The prose gives readers the purpose of the code and the main computational steps.

5.4 Python Example

Code Example 1. The following short Python script applies the explicit finite difference method to the one-dimensional heat equation. The code is included as real text rather than as a screenshot.

```
import numpy as np
import matplotlib.pyplot as plt

# Parameters
alpha = 1.0
x_min = 0.0
x_max = 1.0
final_time = 0.05

num_x = 51
dx = (x_max - x_min) / (num_x - 1)
dt = 0.4 * dx**2 / alpha
num_steps = int(final_time / dt)

r = alpha * dt / dx**2

# Grid and initial condition
x = np.linspace(x_min, x_max, num_x)
u = np.sin(np.pi * x)

# Boundary conditions
u[0] = 0.0
u[-1] = 0.0

# Time stepping
for n in range(num_steps):
    u_new = u.copy()

    for j in range(1, num_x - 1):
        u_new[j] = u[j] + r * (u[j - 1] - 2*u[j] + u[j + 1])

    u_new[0] = 0.0
    u_new[-1] = 0.0
    u = u_new

# Plot the final approximation
```

```
plt.plot(x, u)
plt.xlabel("x")
plt.ylabel("temperature")
plt.title("Approximate solution of the heat equation")
plt.show()
```

The script first defines the physical and numerical parameters. It then creates the spatial grid, applies the initial condition, and advances the solution in time. The inner loop applies the finite difference formula from Equation 5.4 at each interior grid point.

5.5 MATLAB Example

Code Example 2. The following MATLAB script implements the same numerical method. Including both examples is useful in a template because some students work primarily in Python, while others work in MATLAB.

```
% Parameters
alpha = 1.0;
x_min = 0.0;
x_max = 1.0;
final_time = 0.05;

num_x = 51;
dx = (x_max - x_min) / (num_x - 1);
dt = 0.4 * dx^2 / alpha;
num_steps = floor(final_time / dt);

r = alpha * dt / dx^2;

% Grid and initial condition
x = linspace(x_min, x_max, num_x);
u = sin(pi * x);

% Boundary conditions
u(1) = 0.0;
u(end) = 0.0;
```

```
% Time stepping
for n = 1:num_steps
    u_new = u;

    for j = 2:num_x-1
        u_new(j) = u(j) + r * (u(j-1) - 2*u(j) + u(j+1));
    end

    u_new(1) = 0.0;
    u_new(end) = 0.0;
    u = u_new;
end

% Plot the final approximation
plot(x, u, 'LineWidth', 2)
xlabel('x')
ylabel('temperature')
title('Approximate solution of the heat equation')
grid on
```

The MATLAB and Python versions implement the same mathematical method. The indexing differs because MATLAB arrays start at index 1, while Python arrays start at index 0.

5.6 A Small Numerical Summary

Table 5.1 shows a small example of how computational results might be summarized. The table uses real text and column headers rather than a screenshot of program output.

Table 5.1: Sample numerical summary for the explicit heat equation computation.

Grid points	Time step	Stability ratio r	Qualitative behavior
25	$6.94 \cdot 10^{-4}$	0.40	Stable
51	$1.60 \cdot 10^{-4}$	0.40	Stable
101	$4.00 \cdot 10^{-5}$	0.40	Stable

A table like this is usually easier to read than a screenshot of a terminal window. If a table becomes too large, the student should consider using a ‘longtable‘ or moving detailed results to an appendix.

5.7 A Simple Workflow Diagram

Figure 5.1 gives a schematic workflow for a computational experiment. This is the kind of figure that might appear in a computer science, engineering, or data science dissertation.

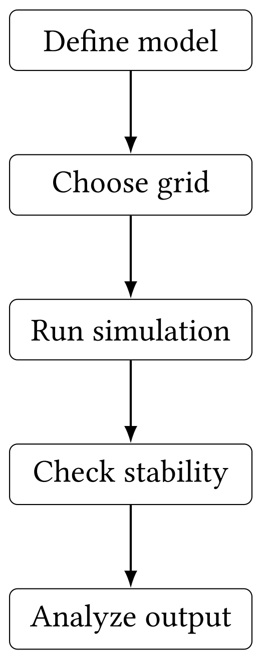

Figure 5.1: A simple workflow for a computational experiment. The researcher defines the model, chooses a grid, runs the simulation, checks stability, and analyzes the output.

5.8 Accessibility Notes for Code and Algorithms

Code should not be included as a screenshot. Screenshots of code are difficult to search, enlarge, copy, or interpret with assistive technology. Instead, use a real text block, and explain the purpose of the code before or after the block.

For accessible computational writing, authors should follow these practices.

1. Introduce the mathematical or computational problem before showing code.

2. Keep code examples short when possible.

3. Explain important variables in prose.
4. Avoid using color alone to communicate meaning in code or figures.
5. Do not paste screenshots of terminal output, tables, or source code.
6. Put long scripts in an appendix or a repository, and include only essential excerpts in the main text.

These habits help readers understand both the computation and the reason it matters.

CHAPTER 6

Conclusions

This chapter summarizes the main findings of the dissertation, discusses the broader conclusions suggested by the work, and identifies possible directions for future study. The goal is to bring together the major themes developed in the earlier chapters and to explain how they contribute to the overall purpose of the research.

6.1 Summary of Results

The results of this dissertation address the central questions introduced at the beginning of the study. Through the methods and analysis presented in the previous chapters, the work identifies several important patterns, relationships, and outcomes. These results provide evidence that the main topic can be understood through a structured approach that connects the background material, the research design, and the interpretation of the findings.

Overall, the study shows that the problem under consideration has meaningful features that can be described, compared, and evaluated. The results also suggest that the framework used in this dissertation is useful for organizing the key ideas and for explaining how the individual parts of the study relate to the broader research question.

6.2 Overarching Conclusions

The conclusions drawn from this work emphasize the importance of careful analysis, clear organization, and thoughtful interpretation. The dissertation demonstrates that the topic is significant not only because of its immediate results, but also because of the larger questions it raises. By examining the problem from multiple perspectives, the study contributes to a more complete understanding of the subject.

A major conclusion is that the ideas developed throughout the dissertation are connected in ways that support the central argument of the work. The findings reinforce the value of the approach taken here and suggest that similar methods may be useful in related contexts. At the same time, the conclusions acknowledge that the study has limits and that additional work is needed to build on these results.

6.3 Future Work

Future work may extend this dissertation in several directions. One possible direction is to apply the methods used here to a broader collection of examples or cases. This would make it possible to test the strength of the conclusions in additional settings and to determine whether the observed patterns remain consistent.

Another direction is to refine the framework introduced in this study. Further research could examine related questions in greater detail, consider alternative methods, or compare the present approach with other approaches. These extensions would help clarify the scope of the results and may lead to new insights about the topic.

Finally, future work may explore practical applications of the findings. By connecting the conclusions of this dissertation to new problems, future studies can continue to develop the ideas presented here and strengthen their usefulness for both research and practice.

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APPENDICES

APPENDIX A

Acronyms

A summary of all acronyms is documented in Table A.1.

Table A.1: A summary of acronyms used in alphabetical order.

Acronym	Abbreviation
African Easterly Wave	AEW
African Easterly Jet	AEJ
African Monsoon Multidisciplinary Analysis	AMMA
Climate Forecasting System Reanalysis	CFSR
Convective Available Potential Energy	CAPE
Dry Adiabatic Lapse Rate	DALR
European Centre for Medium Range Weather Forecasting	ECMWF
Eddy Kinetic Energy	EKE
GARP Atlantic Tropical Experiment	GATE
Global Forecast System	GFS
Global Precipitation Mission	GPM
Long Wave (Radiation)	LW
Mesoscale Convective System	MCS
National Aeronautical and Space Administration	NASA
National Center for Atmospheric Research	NCAR
National Center for Environmental Prediction	NCEP
National Hurricane Center	NHC
National Science Foundation	NSF
Planetary Boundary Layer	PBL

Potential Vorticity	PV
Short Wave (Radiation)	SW
Quasi-Geostrophic	QG
Rossby Wave	RW
Tropical Cyclone	TC
TRMM Multi-Satellite Precipitation Analysis	TMPA
Tropical Rainfall Measurement Mission	TRMM
Weather Research and Forecasting Model	WRF

APPENDIX B

Variables

A summary of all variables is documented in Table B.1.

Table B.1: A summary of common meteorological variables and their abbreviations in alphabetical order.

Variable	Abbreviation
Arbitrary variable	X
Absolute vorticity vector	$\vec{\eta}$
Coriolis parameter	f
Gas constant for dry air	R
Geopotential	Φ
Geopotential height	Z
Gravitational accelleration	g
Horizontal wind vector	$\vec{V}$
Isobaric vertical motion	ω
Latent heat of vaporization	l_v
Meridional unit vector	$\hat{j}$
Meridional wind	v
Potential temperature	θ
Potential vorticity	P
Pressure	p
Relative vorticity vector	$\vec{\zeta}$
Specific density	α
Specific heat capacity at constant pressure	c_p

Temperature	T
Time	t
Three-dimensional wind vector	$\vec{U}$
Vertical absolute vorticity	η
Vertical unit vector	$\hat{k}$
Vertical relative vorticity	ζ
Zonal unit vector	$\hat{\imath}$
Zonal wind	u